

E-commerce Marketing Funnel Analytics

Data Dictionary and DAX Measures Documentation

Purpose: This document explains the final Power BI data model, table definitions, field meanings, relationship structure, and DAX measures used in the e-commerce customer journey and conversion funnel analytics dashboard.

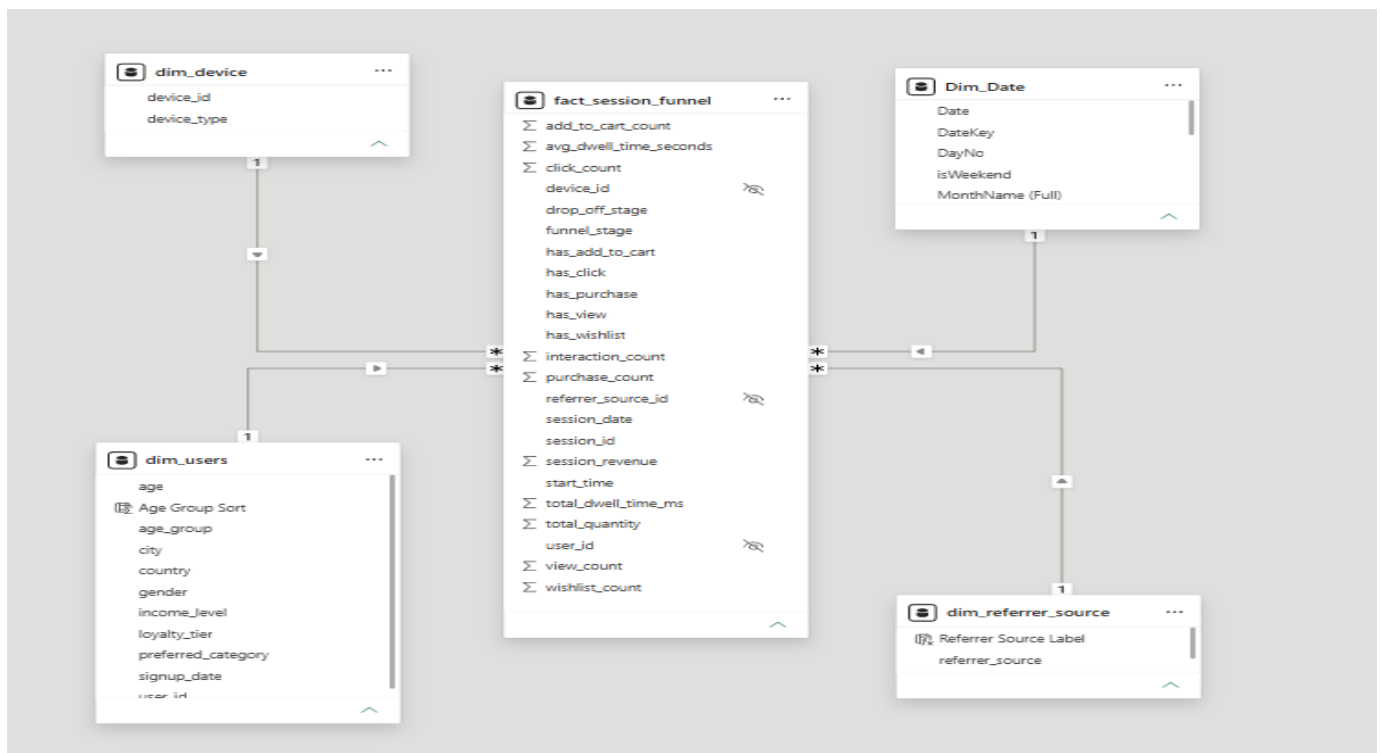
Business question: Which traffic sources, devices, and customer segments drive the highest conversion, and where do users drop off before purchase?

1. Model Summary

The reporting model uses a simplified star schema with one main session-level fact table and supporting dimensions. The final model is intentionally focused on marketing funnel analysis rather than full product catalogue or review analysis.

Table	Type	Grain / Level	Main Purpose
fact_session_funnel	Fact	One row per session	Tracks session-level funnel behaviour, interaction counts, purchases, revenue, funnel stage, and drop-off stage.
dim_users	Dimension	One row per user	Provides customer attributes for segmentation by age group, income level, loyalty tier, location, gender, and preferred category.
dim_device	Dimension	One row per device type	Supports analysis by device type such as desktop, mobile, and tablet.
dim_referrer_source	Dimension	One row per traffic source	Supports channel analysis by referrer source, with cleaned display labels for dashboard visuals.
Dim_Date	Dimension	One row per calendar date	Created in Power BI to support time-based filtering, Australian locale formatting, and monthly trend analysis.

Power BI Model Schema



2. Relationship Structure

The model uses one-to-many relationships from dimensions to the fact table. The recommended Power BI relationship direction is single direction from dimension to fact.

From Table	From Column	To Table	To Column	Cardinality	Filter Direction
dim_users	user_id	fact_session_funnel	user_id	1:*	Single
dim_device	device_id	fact_session_funnel	device_id	1:*	Single
dim_referrer_source	referrer_source_id	fact_session_funnel	referrer_source_id	1:*	Single
Dim_Date	Date	fact_session_funnel	session_date	1:*	Single

3. Table Data Dictionary

fact_session_funnel

Fact table - one row per customer session. This table is the central table used for funnel, conversion, revenue, source, device, and drop-off analysis.

Column	Data Type	Description	Example / Notes
session_id	Text	Unique identifier for each session.	Primary key of the fact table.
user_id	Text	User associated with the session.	Foreign key to dim_users.
session_date	Date	Date of the session.	Connected to Dim_Date[Date].
device_id	Whole number	Device key for the session.	Foreign key to dim_device.
referrer_source_id	Whole number	Traffic source key for the session.	Foreign key to dim_referrer_source.
start_time	Date/Time	Session start timestamp.	Used for validation or detailed analysis.
has_view	Boolean	Indicates whether the session included a view interaction.	TRUE/FALSE.
has_click	Boolean	Indicates whether the session included a click interaction.	TRUE/FALSE.
has_add_to_cart	Boolean	Indicates whether the session included an add-to-cart interaction.	TRUE/FALSE.
has_wishlist	Boolean	Indicates whether the session included wishlist behaviour.	TRUE/FALSE.
has_purchase	Boolean	Indicates whether the session resulted in a purchase.	Used to calculate converted sessions.
view_count	Whole number	Number of view interactions in the session.	Aggregated from interactions.
click_count	Whole number	Number of click interactions in the session.	Aggregated from interactions.
add_to_cart_count	Whole number	Number of add-to-cart interactions in the session.	Aggregated from interactions.
wishlist_count	Whole number	Number of wishlist interactions in the session.	Aggregated from interactions.
interaction_count	Whole number	Total interactions within the session.	View + click + cart + wishlist and other valid interactions.
total_dwell_time_ms	Whole number	Total dwell time across session interactions in milliseconds.	Used for engagement analysis.
avg_dwell_time_seconds	Decimal	Average dwell time per interaction in seconds.	Converted from milliseconds.
purchase_count	Whole number	Number of purchases in the session.	Used in purchase and AOV metrics.
total_quantity	Whole number	Total quantity purchased in the session.	Aggregated from purchases.
session_revenue	Decimal / Currency	Revenue generated by the session.	Used in revenue and revenue per session measures.
funnel_stage	Text	Highest funnel stage reached by the session.	Viewed, Clicked, Added to Cart, Purchased.
drop_off_stage	Text	Stage where the session ended before purchase, or Converted.	Drop after View, Drop after Click, Drop after Cart, Converted.

dim_users

Customer dimension used for segmentation and targeting analysis.

Column	Data Type	Description	Example / Notes
user_id	Text	Unique customer identifier.	Primary key.
age	Whole number	Customer age.	Used to create age group.
Age Group Sort	Whole number	Sort key for age group display.	1 = 18-24, 2 = 25-34, etc.
age_group	Text	Grouped age band for dashboard analysis.	18-24, 25-34, 35-44, 45-54, 55+.
gender	Text	Customer gender field from the source data.	Used as optional segment field.
country	Text	Customer country.	Used for location-based segmentation.
city	Text	Customer city.	Optional location analysis.
signup_date	Date	Date when the user signed up.	Can be used for cohort or tenure analysis.
income_level	Text	Customer income segment.	low, medium, high, very_high.
preferred_category	Text	Customer preferred product category.	Used as optional segment context.
loyalty_tier	Text	Customer loyalty tier.	bronze, silver, gold, platinum.

dim_device

Device dimension used to compare behaviour and conversion across device types.

Column	Data Type	Description	Example / Notes
device_id	Whole number	Unique device type key.	Primary key.
device_type	Text	Device category used during the session.	desktop, mobile, tablet.

dim_referrer_source

Traffic source dimension used for channel performance and drop-off analysis.

Column	Data Type	Description	Example / Notes
referrer_source_id	Whole number	Unique traffic source key.	Primary key.
referrer_source	Text	Original cleaned traffic source.	organic_search, social_media, paid_search, display_ad, etc.
Referrer Source Label	Text	Short display label for dashboard visuals.	Organic, Social, Paid, Display, Direct, Email, Referral.

Dim_Date

Date dimension created in Power BI to support date filtering and time intelligence.

Column	Data Type	Description	Example / Notes
Date	Date	Calendar date.	Primary date column marked as the date table.
DateKey	Whole number	Numeric date key.	YYYYMMDD format.
DayNo	Whole number	Day number in month.	1-31.
isWeekend	Boolean	Indicates whether the date is a weekend.	TRUE/FALSE.
MonthName (Full)	Text	Full month name.	January, February, etc.
YearMonth	Text	Year-month label for trends.	2025-01.
YearMonthSort	Whole number	Sort key for YearMonth.	YYYYMM.

4. DAX Measures Dictionary

The dashboard uses dedicated measure tables to keep DAX measures organised by analytical purpose. The measures below are the key calculations used across the four Power BI pages.

Core Measures

Measure	Purpose	DAX Formula
Total Sessions	Counts distinct sessions in the current filter context.	Total Sessions = DISTINCTCOUNT (fact_session_funnel[session_id])
Total Users	Counts distinct users in the current filter context.	Total Users = DISTINCTCOUNT (fact_session_funnel[user_id])
Total Revenue	Calculates total revenue from all sessions.	Total Revenue = SUM (fact_session_funnel[session_revenue])
Total Purchases	Calculates total purchase count.	Total Purchases = SUM (fact_session_funnel[purchase_count])
Total Quantity	Calculates total quantity purchased.	Total Quantity = SUM (fact_session_funnel[total_quantity])
Average Order Value	Average revenue per purchase.	Average Order Value = DIVIDE ([Total Revenue], [Total Purchases])
Revenue per Session	Average revenue generated per session.	Revenue per Session = DIVIDE ([Total Revenue], [Total Sessions])
Revenue per User	Average revenue generated per user.	Revenue per User = DIVIDE ([Total Revenue], [Total Users])

Funnel Step Measures

Measure	Purpose	DAX Formula
Sessions with View	Counts sessions that reached the view stage.	Sessions with View = CALCULATE ([Total Sessions], fact_session_funnel[has_view] = TRUE())
Sessions with Click	Counts sessions that reached the click stage.	Sessions with Click = CALCULATE ([Total Sessions], fact_session_funnel[has_click] = TRUE())
Sessions with Add to Cart	Counts sessions that reached the add-to-cart stage.	Sessions with Add to Cart = CALCULATE ([Total Sessions], fact_session_funnel[has_add_to_cart] = TRUE())
Sessions with Wishlist	Counts sessions that included wishlist behaviour.	Sessions with Wishlist = CALCULATE ([Total Sessions], fact_session_funnel[has_wishlist] = TRUE())
Converted Sessions	Counts sessions that resulted in a purchase.	Converted Sessions = CALCULATE ([Total Sessions], fact_session_funnel[has_purchase] = TRUE())

Conversion Rate Measures

Measure	Purpose	DAX Formula
Purchase Conversion Rate	Percentage of sessions that resulted in purchase.	Purchase Conversion Rate = DIVIDE ([Converted Sessions], [Total Sessions])
View Rate	Share of sessions with at least one view.	View Rate = DIVIDE ([Sessions with View], [Total Sessions])

Measure	Purpose	DAX Formula
Click Rate	Share of sessions with at least one click.	Click Rate = DIVIDE ([Sessions with Click], [Total Sessions])
Add to Cart Rate	Share of sessions with add-to-cart behaviour.	Add to Cart Rate = DIVIDE ([Sessions with Add to Cart], [Total Sessions])
Wishlist Rate	Share of sessions with wishlist behaviour.	Wishlist Rate = DIVIDE ([Sessions with Wishlist], [Total Sessions])

Funnel Transition Rates

Measure	Purpose	DAX Formula
View to Click Rate	Measures how many view sessions progressed to click.	View to Click Rate = DIVIDE ([Sessions with Click], [Sessions with View])
Click to Cart Rate	Measures how many click sessions progressed to add-to-cart.	Click to Cart Rate = DIVIDE ([Sessions with Add to Cart], [Sessions with Click])
Cart to Purchase Rate	Measures how many add-to-cart sessions became purchases.	Cart to Purchase Rate = DIVIDE ([Converted Sessions], [Sessions with Add to Cart])
View to Purchase Rate	Measures purchase conversion among sessions that reached view.	View to Purchase Rate = DIVIDE ([Converted Sessions], [Sessions with View])
Funnel Transition Rate	Returns the correct transition rate for the selected transition step.	Funnel Transition Rate = SWITCH (SELECTEDVALUE ('Funnel Transition Steps'[Transition]), "View → Click", [View to Click Rate], "Click → Cart", [Click to Cart Rate], "Cart → Purchase", [Cart to Purchase Rate])

Drop-off Measures

Measure	Purpose	DAX Formula
View to Click Drop-off	Number of sessions lost between view and click.	View to Click Drop-off = [Sessions with View] - [Sessions with Click]
Click to Cart Drop-off	Number of sessions lost between click and add-to-cart.	Click to Cart Drop-off = [Sessions with Click] - [Sessions with Add to Cart]
Cart to Purchase Drop-off	Number of sessions lost between cart and purchase.	Cart to Purchase Drop-off = [Sessions with Add to Cart] - [Converted Sessions]
Drop-off Share	Share of sessions within a selected drop-off stage.	Drop-off Share = DIVIDE ([Total Sessions], CALCULATE ([Total Sessions], ALLSELECTED (fact_session_funnel[drop_off_stage])))
Non-converted Sessions	Sessions that did not result in purchase.	Non-converted Sessions = [Total Sessions] - [Converted Sessions]
Non-conversion Rate	Share of sessions that did not convert.	Non-conversion Rate = DIVIDE ([Non-converted Sessions], [Total Sessions])

Interaction Measures

Measure	Purpose	DAX Formula
Total Interactions	Total interaction count across sessions.	Total Interactions = SUM (fact_session_funnel[interaction_count])
Total Views	Total view interactions.	Total Views = SUM (fact_session_funnel[view_count])
Total Clicks	Total click interactions.	Total Clicks = SUM (fact_session_funnel[click_count])
Total Add to Cart	Total add-to-cart interactions.	Total Add to Cart = SUM (fact_session_funnel[add_to_cart_count])

Measure	Purpose	DAX Formula
Total Wishlist	Total wishlist interactions.	Total Wishlist = SUM (fact_session_funnel[wishlist_count])
Interactions per Session	Average number of interactions per session.	Interactions per Session = DIVIDE ([Total Interactions], [Total Sessions])
Average Dwell Time Seconds	Average dwell time per interaction/session record.	Average Dwell Time Seconds = AVERAGE (fact_session_funnel[avg_dwell_time_seconds])

Time Intelligence Measures

Measure	Purpose	DAX Formula
Revenue Previous Month	Revenue from the previous month.	Revenue Previous Month = CALCULATE ([Total Revenue], PREVIOUSMONTH (Dim_Date[Date]))
Revenue MoM Change	Month-over-month revenue change.	Revenue MoM Change = [Total Revenue] - [Revenue Previous Month]
Revenue MoM Change %	Month-over-month revenue percentage change.	Revenue MoM Change % = DIVIDE ([Revenue MoM Change], [Revenue Previous Month])
Sessions Previous Month	Sessions from the previous month.	Sessions Previous Month = CALCULATE ([Total Sessions], PREVIOUSMONTH (Dim_Date[Date]))
Sessions MoM Change %	Month-over-month sessions percentage change.	Sessions MoM Change % = DIVIDE ([Sessions MoM Change], [Sessions Previous Month])
Conversion Rate Previous Month	Conversion rate from the previous month.	Conversion Rate Previous Month = CALCULATE ([Purchase Conversion Rate], PREVIOUSMONTH (Dim_Date[Date]))

Dashboard Helper Measures

Measure	Purpose	DAX Formula
Selected Date Range	Dynamic text measure showing the selected date range.	Selected Date Range = VAR MinDate = MIN (Dim_Date[Date]) VAR MaxDate = MAX (Dim_Date[Date]) RETURN FORMAT (MinDate, "dd mmm yyyy") & " - " & FORMAT (MaxDate, "dd mmm yyyy")
Selected Referrer Source	Dynamic text measure showing selected traffic sources.	Selected Referrer Source = IF (ISFILTERED (dim_referrer_source[referrer_source]), CONCATENATEX (VALUES (dim_referrer_source[referrer_source]), dim_referrer_source[referrer_source], ", "), "All Sources")
Selected Device Type	Dynamic text measure showing selected device types.	Selected Device Type = IF (ISFILTERED (dim_device[device_type]), CONCATENATEX (VALUES (dim_device[device_type]), dim_device[device_type], ", "), "All Devices")

5. Dashboard Page Mapping

The measures and fields are used across four Power BI pages. Each page answers a different level of the business question.

Page	Main Question	Key Fields / Measures
Executive Overview	How is the overall funnel performing?	Total Sessions, Total Users, Total Revenue, Purchase Conversion Rate, Funnel Sessions, Drop-off Stage, Referrer Source.
Channel Performance	Which traffic sources and devices drive better conversion and revenue quality?	Referrer Source, Device Type, Total Sessions, Purchase Conversion Rate, Revenue per Session, Total Revenue.
Customer Segment Analysis	Which customer segments should be prioritised for targeting?	Loyalty Tier, Income Level, Age Group, Revenue per Session, Revenue per User, Purchase Conversion Rate, Source & Device Heatmap.
Conversion Opportunities	Where are the biggest conversion opportunities and what should the business do next?	Drop-off Stage, Funnel Transition Rate, View to Click Rate, Click to Cart Rate, Cart to Purchase Rate, Drop-off Heatmap.

6. Insight & Action Guide

This guide is designed to remain valid when the user changes filters in Power BI. It provides an action framework based on the selected drop-off stage, weakest transition, and source-level drop-off pattern.

Funnel Issue	What it Means	Recommended Actions
Drop after View	Users viewed products or pages but did not continue to click.	Improve landing page relevance, make value proposition clearer, review page speed and images, and strengthen call-to-action visibility.
Drop after Click	Users clicked but did not move to cart.	Review product detail pages, pricing clarity, offer presentation, trust signals, comparison information, and CTA placement.
Drop after Cart	Users added items to cart but did not complete purchase.	Prioritise cart abandonment recovery, review checkout friction, improve delivery/payment transparency, and consider limited-time offers.
Weak Cart to Purchase Rate	The final conversion step is underperforming.	Investigate checkout barriers, unexpected fees, payment errors, slow checkout, mobile issues, and test simplified checkout or guest checkout.
Source-level Drop-off	Some traffic sources contribute more heavily to specific drop-off stages.	Prioritise sources with high drop-off volume, review landing page relevance, focus retargeting on cart drop-off, and compare source performance with revenue per session.

7. Recommended Formatting

Measure / Field Type	Recommended Format
Revenue measures	Currency, Australian dollar formatting where appropriate.
Conversion and transition rates	Percentage with 1-2 decimal places.
Session, user, purchase, and interaction counts	Whole number with display units where useful.
Revenue per session and revenue per user	Currency with 2 decimal places.
Heatmap conditional formatting	Low: #F7EFE8, Middle: #E8C98E, High: #0E1E2F or #2F5964 if text contrast is needed.